Racially Divergent Patterns in the Relationship between Same-Race Subordinates and Federal Supervisors' Job Satisfaction

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Abstract

Despite substantial evidence on how supervisors' racial composition influences employees' work-related attitudes, little attention has been given to the reciprocal effects on supervisors themselves. This study examines how same-race subordinates in one's agency affect supervisors' job satisfaction, revealing a racially divergent pattern. Using federal employee data from 2020–2023, this paper finds that White supervisors experience increased job satisfaction when working in agencies with higher proportions of same-race subordinates, consistent with relational demography theory. In contrast, supervisors of color experience decreased job satisfaction under similar conditions, aligning with theoretical expectations regarding the burden of representation and tokenism theory. Mediation analysis demonstrates that this negative association operates through reduced role clarity and deteriorated supervisor relationships, consistent with theory arguments that supervisors of color may face role conflicts between organizational norms and community expectations. However, this burden diminishes as minority representation in leadership increases, becoming positive when exceeding 44% of supervisors. These findings reveal hidden costs borne by minority supervisors and identify when these costs transform into benefits.

Keywords—diversity, job satisfaction, workplace attitudes and behaviors, representative bureaucracy, Federal government

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1. INTRODUCTION

A substantial body of literature demonstrates that when supervisors' demographics (race, ethnicity, and gender) match those of their subordinates, employees experience improved job satisfaction, whether measured at the dyadic level (Grissom & Keiser, 2011; Grissom et al., 2012) or at the agency level (Moon & Jung, 2018). However, little attention has been given to the reciprocal impact on supervisors themselves. This gap is particularly important because existing theories suggest inconsistent expectations about how subordinates' demographic composition might affect supervisors' well-being, depending on their racial group.

On the one hand, relational demography theory suggests that individuals benefit from working with people from similar backgrounds due to shared knowledge, culture, and improved communication (Tsui et al., 1992; Tsui & O'Reilly III, 1989). Applied to supervisor-subordinate relationships, this theory predicts that working in workplaces with more subordinates from similar racial/ethnic backgrounds should enhance supervisors' job satisfaction through smoother cooperation and increased organizational commitment (Hennessy & West, 1999; Mueller et al., 1999; Turner, 2010). The underlying mechanism is that shared backgrounds facilitate easier communication and mutual understanding, creating reciprocal benefits for both parties.

On the other hand, the burden of representation and tokenism literature suggests the opposite dynamic may occur for supervisors of color. Research on minority bureaucrats reveals that they encounter significant challenges balancing expectations from their community with organizational norms (Carroll et al., 2019; Habyarimana et al., 2007; Herbert, 1974). In particular, supervisors are embedded in organizational hierarchies, operating at middle levels where they must navigate expectations from both their subordinates and higher-level leadership or political appointees. Subordinates might expect the same-race supervisors in their agency to "represent"

their voices and advocate for their rights, such as supporting equity initiatives. However, minority supervisors may encounter conflict and tension when such advocacy conflicts with organizational priorities, resource constraints, or expectations from the higher authority (Lee, 2022; Miller et al., 2018). Additionally, tokenism theory suggests that minoritized leaders experience isolation and pressure due to their numerical minority status, being excluded from dominant group support networks while facing intensified performance pressures and being perceived as representatives of their racial category rather than as individuals (Kanter, 1977a; 1977b; Cook & Glass, 2015; Moore, 2013). These dimensions collectively produce strain that harms job satisfaction and well-being.

These competing theoretical perspectives raise fundamental questions about whether—and for whom—same-race subordinates enhance or diminish supervisor well-being. This study addresses three key research questions: (1) How does having more same-race subordinates in their agency affect supervisors' job satisfaction? (2) Does this association vary depending on the racial background of the supervisors? (3) Under what conditions are negative associations mitigated or transformed into benefits?

To investigate these questions, this study uses federal employee data in the United States from 2020 to 2023 and examines 403,135 supervisory personnel across 32 agencies. The analysis reveals a racially divergent relationship: while White supervisors experienced an increased job satisfaction when having more same-race subordinates in their agency, supervisors of color experienced decreased job satisfaction. Through mediation analysis, this paper identifies that role clarity and supervisor relationships account for the negative association of racial congruence for supervisors of color. Critically, the present study demonstrates that these negative associations diminish and become positive when minority representation in leadership reaches approximately 44%, providing empirical support for critical mass theory.

This paper makes several crucial contributions to the theory and practice of human resources management in the public sector. First, this paper identifies the racially divergent relationships between same-race subordinate composition at the agency level and supervisors' well-being. The findings reveal that while White supervisors benefit from working in an agency with more same-race subordinates (as predicted by the relational demography theory), supervisors of color experienced negative consequences (as expected by the burden of representation theory). However, this association is conditional on the numeric distribution of supervisor composition at the leadership level. When supervisors of color gain greater access to leadership, they will be better able to navigate challenging situations. These findings suggest that, while relational demography theory and the burden of representation literature make distinct predictions about the relationship, they are not necessarily in conflict; rather, the two may yield different patterns depending on the racial composition within organizations.

Second, this paper also contributes by generating insights into the (lack of) reciprocal effects of demographic congruence between supervisor and agency-level subordinate composition. While past literature concludes that minority subordinates benefit from having supervisors who match their demographics or from working in agencies that have a greater share of leadership aligned with their demographics (Grissom et al., 2012; Grissom & Keiser, 2011; Moon & Jung, 2018), this paper shows that the patterns are reversed for supervisors of color, at least until the percent of minority supervisors passes a threshold.

Moreover, while promoting people of color to leadership has been viewed as a solution to improve diversity management and benefit employees from minoritized backgrounds (Choi, 2009; Moon & Jung, 2018), it is important to consider the additional challenges that minority supervisors themselves may face. By revealing the hidden costs and identifying when they transform into

benefits, this research provides a roadmap for more effective and equitable diversity initiatives in public organizations.

2. LITERATURE REVIEW AND HYPOTHESES DEVELOPMENT

2.1. The Benefits of Working with People from Similar Backgrounds

Relational demography theory indicates that individuals tend to have more positive experiences when interacting with others from similar demographic backgrounds (Tsui et al., 1992). Previous studies have shown that compatibility and demographic similarity among coworkers are linked to higher organizational commitment, job satisfaction, and lower turnover intentions (Avery et al., 2007; Choi, 2017; Hauret & Williams, 2020; Hoppe et al., 2014; Mueller et al., 1999; Nielsen & Madsen, 2017; Stainback & Irvin, 2012; Wang & Brower, 2019). Two key mechanisms explain these patterns.

First, similarity-attraction theory suggests people with similar backgrounds share knowledge and culture, leading to better cooperation and communication (Habyarimana et al., 2007; Tsui et al., 1992). For example, studies find that demographic congruence increases knowledge sharing and coworker support (Lin, 2007; Mueller et al., 1999). Second, the social identity perspective posits that interacting with similar others reinforces identity and belongingness, increasing cohesion and organizational commitment (Hennessy & West, 1999; Mueller et al., 1999; Turner, 2010). Previous studies have found that people who encounter a higher proportion of colleagues with similar demographic backgrounds or values are likely to experience a higher level of cohesion, leading to a commitment to the organization (Chen et al., 1998; Gonzalez, 2016; Mueller et al., 1999; Paillé, 2013; Popielarz & McPherson, 1995).

This relationship also applies to the interactions between supervisors and subordinates. For example, Bakar and McCann (2014) found that ethnic and gender congruence between supervisors

and subordinates in Malaysia improved perceptions of the relationship for both parties. Jackson and Johnson (2012) found that in the US, similarity in self-identity between supervisors and subordinates led to better relationships, as reported by either supervisors or subordinates. This positive relationship could then lead to higher job satisfaction for supervisors (Avery et al., 2007; Hauret & Williams, 2020; Hoppe et al., 2014; Nielsen & Madsen, 2017; Stainback & Irvin, 2012; Wang & Brower, 2019).

2.2. The Burdens of Representation

Despite the benefits of working with people from similar demographic backgrounds, underrepresented managers also struggle with the tension between organizational expectations, their racial identity, and the expectations of their same-race subordinates (Arifeen & Syed, 2019; Carroll et al., 2019; Simons et al., 2007). Studies in the public sector have shown that bureaucrats from minority backgrounds often face tensions between organizational goals and the expectations of their own communities (Althaus & O'Faircheallaigh, 2022; Herbert, 1974; Carroll et al., 2019; Feit, 2024; Watkins-Hayes, 2011). For instance, Watkins-Hayes (2011) interviewed Black and Latino bureaucrats in welfare agencies, revealing that these individuals were caught between conflicting expectations: those of their clients, who expected service delivery in a certain manner based on shared demographic backgrounds, and those of the organization, which demanded adherence to rules and procedures. This tension places minority bureaucrats under significant pressure. Furthermore, Carroll et al. (2019) note that as bureaucrats ascend the hierarchy, the tension between organizational norms and racial identity intensifies as both the community and the organization place higher expectations on them.

Still, past studies regarding the burden of representation focus on bureaucrats and their clients and have not yet explored whether similar tension arises between supervisors and their

subordinates. The pattern observed in bureaucrat-client relationships might also apply to supervisor-subordinate interactions. The central idea of representative bureaucracy theory, as well as the burden of representation literature, is that shared demographic backgrounds (which imply understanding and belonging) change how bureaucrats exercise discretion in decision-making and policy implementation (Marvel & Resh, 2015; Meier & Bohte, 2001; Tummers & Bekkers, 2014). In this sense, supervisors managing their subordinates face situations similar to bureaucrats serving their clients, as both involve discretion and decision-making that may favor or oppose the interests of their subordinates/clients. It follows that minority supervisors may face challenges when their desire to advocate for their employees' rights and benefits conflicts with the organization's procedures, norms, or goals. For example, based on representative bureaucracy theory, minority employees might expect their same-race supervisors to show greater understanding of their unique challenges and provide advocacy or support for minority interests within the agency (Simons et al., 2007; White-Davis et al., 2016). However, supervisors may face organizational pressures that prevent them from implementing personnel practices that benefit their own-race subordinates (Lee, 2022). They might even feel compelled to impose more stringent demands on these employees to maintain their organizational status and to ensure that an underperforming own-race subordinate does not negatively affect their leadership (Miller et al., 2018).

Literature focusing on the private sector offers complementary insights. Previous studies found that minority and female managers experience role conflict when the mainstream norm diverges from the expectations of their social group (Arifeen & Syed, 2019; Miller et al., 2018). For example, Miller et al. (2018) analyzed data from the private sector and found that minority supervisors with higher racial similarity to their intermediate subordinates experienced more relationship conflict and less trust. While the private sector context may differ greatly from the

public sector, the fact that public sectors typically have strict hierarchies and inflexible human resource management rules (Boyne et al., 1999; Williamson, 2011) may intensify tensions for minority supervisors advocating for their same-race subordinates. Additionally, public organizations typically face more conflicting values and greater goal ambiguity (Pandey & Rainey, 2006; Rainey & Jung, 2015). On the one hand, supervisors are subject to complex organizational goals and regulations; on the other hand, public organizations bear expectations to serve as ideal employers and maintain equity in their work environments (Boyne et al., 1999). These pressures may place minority supervisors under greater tension.

Drawing on relational demography theory and the burden of representation literature, this paper argues that same-race subordinate composition has racially divergent patterns. This paper expects that White supervisors experience positive outcomes on their job satisfaction when exposed to greater shares of same-race subordinates, while supervisors of color experience the opposite (decreased job satisfaction). The logic of these racially divergent patterns is that White supervisors benefit from working with people who share similar values—as relational demography theory predicts—without experiencing the potential role conflict suggested by the burden of representation. Conversely, supervisors of color may struggle with the tensions created by feeling representation burdens, which outweigh any benefits from demographic similarity.

While both theoretical perspectives were originally developed to explain interpersonal dynamics, their logic extends to organizational-level demographic composition. Previous research has demonstrated that relational demography operates at both the dyadic level (Bakar & McCann, 2014) and the aggregate organizational level (Mueller et al., 1999; Wharton et al., 2000), as the central premise is that people feel more comfortable in environments with a greater share of similar others. For instance, racial congruence benefits subordinates both in immediate supervisor-

subordinate dyads (Grissom & Keiser, 2011) and when measured through agency-level representation indices (Moon & Jung, 2018). Similarly, the burden of representation operates at the aggregate level because agency-level demographic composition not only influences the likelihood of supervising same-race subordinates but also shapes the organizational environment and expectations for advocacy (Choi, 2013, 2017). Therefore, this paper hypothesizes:

*H_1: White supervisors tend to have **higher** job satisfaction when they work in agencies with a higher proportion of subordinates of the same race/ethnicity.*

*H_2: Supervisors of minority backgrounds tend to have **lower** job satisfaction when they work in agencies with a higher proportion of subordinates of the same race/ethnicity.*

2.3. The Mechanisms of the Burdens

The literature suggests various mechanisms through which subordinates' demographic similarity influences supervisors' job satisfaction. Based on the availability of measurements in the FEVS dataset, this study focuses particularly on examining the mechanisms behind H_2, which proposes potentially counteracting effects for minority supervisors. The burden of representation literature suggests two pathways through which subordinates' composition could negatively affect supervisors' well-being.

The first pathway relates to supervisors' internal tension arising from role conflict. The key argument of the burden of representation theory is that bureaucrats face a tension between their advocacy roles for their own community and their role as officials, requiring them to comply with organizational norms. When one assumes two roles (advocate for the minority community and defender of the organization's goals as defined by the majority), potential conflicts may arise, hindering role clarity.

Studies argue that role incongruence and conflict are major sources of tension faced by minority bureaucrats (Carroll et al., 2019; Habyarimana et al., 2007; Herbert, 1974). Research also demonstrates that bureaucrats face conflicting expectations between their organization and the groups they serve or interact with (Satheesh et al., 2024). In particular, minority managers may feel a commitment to their own community and wish to advocate for individuals of their own background. However, organizational procedures and norms may restrict their ability to take meaningful action. In other words, supervisors of color face two role expectations: on one hand, they serve as "representatives" of their own groups and are expected to advocate for minorities' interests; on the other hand, they are also managers of the organization who should follow organizational rules and pursue agency goals, which may not align with their community's expectations. This conflict diminishes minority supervisors' well-being. Specifically, as the proportion of same-race subordinates in an agency increased, tension due to role conflict may escalate. Role conflict can be operationalized as a lack of role clarity (Hassan, 2013; Jong, 2016); therefore:

*H_3: A higher proportion of subordinates of the same race/ethnicity in the agency is associated with a **lower** level of role clarity, which **reduces** minority supervisors' job satisfaction. (Role clarity **mediates** H_2.)*

The second mechanism involves interactions and relationships with their managers at the higher hierarchy. The key argument from the burden of representation literature is that minority bureaucrats may encounter challenges within the organization when advocating for the interests of the community they represent. That is, serving in advocacy roles may create tension between the bureaucrats and the organization they work for, particularly with organizational leadership.

Specifically, when minority supervisors seek to advocate for their same-race subordinates, but their values conflict with organizational norms, tension could arise in interactions with the organization's leadership. Althaus and O'Faircheallaigh (2022) found that Indigenous bureaucrats who advocate for Indigenous groups may be labeled as difficult by the organization. Additionally, Herbert (1974) and Carroll et al. (2019) highlight colleague pressure and organizational socialization as crucial sources of dilemma for minority administrators. Therefore, as the proportion of same-race subordinates in the agency increases, supervisors of color may experience a worsened relationship with agency leadership due to increased tension between an advocacy role and organizational norms, and/or supervisors may feel more conflicted over their roles. This impact would then lead to lower job satisfaction. Formally, this paper proposes that:

*H_4: A higher proportion of subordinates of the same race/ethnicity in the agency is associated with a **worse** quality of supervisor relationships, which **reduces** minority supervisors' job satisfaction. (Supervisor relationship **mediates** H_2.)*

2.4. From Tokenism to Critical Mass

While the previous section explored potential mechanisms through which same-race subordinates might affect minority supervisors' job satisfaction, these patterns may not be uniform across all organizational contexts. An important contextual factor that could moderate this relationship is the numerical representation of the supervisor's racial/ethnic group within the organization, particularly at leadership levels. The theories of tokenism and critical mass provide valuable insights into how the proportion of same-race colleagues might influence the experiences of minority supervisors, potentially altering the relationship established in the previous hypotheses.

Kanter (1977a, 1977b) highlights how numerical distribution within an organization shapes the status of minoritized groups. She discusses how women in skewed organizations, where

females account for a very small share, suffer from being treated as a "token." As tokens, these women are viewed as symbols or representatives of their categories rather than as individuals, leading to over-pressure, isolation, and encapsulation in stereotypical roles. While Kanter's original theory focuses on gender effects in private corporations, subsequent studies have also identified similar patterns for gender minorities in the public sector (Choi & Ko, 2025; Gidengil & Vengroff, 1997; Harris, 2013) and for racial/ethnic minorities in various settings (Binderkrantz, 2011; Cook & Glass, 2015; Moore, 2013; Stroshine & Brandl, 2011). The central argument of tokenism theory is that the negative experiences of minority groups in the workplace are not solely due to their social status but are also attributed to numerical distribution within the organization. The more skewed the composition, the more likely they are to experience the negative effects.

However, this adverse situation might be mitigated once the minority group occupies a significant share in the working environment. In particular, as minorities gain more representation in the organization at the leadership level, supervisors of color would have more colleagues to support the communities they represent, which could help mitigate the tension with the organization. While studies vary on the threshold for a critical mass, the consensus is that the size of the minority matters (Cook & Glass, 2015; Fredette & Bernstein, 2019; Harris, 2013; Li, 2024). For example, Jackson et al. (1995) found that the level of depression and role overload among African Americans in leadership positions decreased once the workplace composition moved from being dominated by Whites to a more balanced environment. Kalbfeld (2019) also finds that the feeling of isolation experienced by minority students is reduced once they obtain a higher share on campus. Therefore, this paper expects that:

H_5: As the proportion of same-race/ethnicity colleagues at the leadership level increases, the negative effects of the proportion of same-race/ethnicity subordinates would reduce. (The proportion of colleagues of the same race/ethnicity ***moderates*** *H_2.)*

2.5. Theoretical Framework

Figure 1 summarizes the theoretical framework and hypotheses. In the model, job satisfaction is treated as the ultimate outcome, while role clarity and supervisor relationships are mediators. The key independent variable is the demographic similarity between supervisors and their subordinates at the agency level. The racial composition of colleagues at the leadership level is treated as a moderating variable.

[Figure 1 Inserted Here]

3. DATA AND METHOD

3.1. Data and Sample

The data employed in this paper comprises the Federal Employee Viewpoint Survey (FEVS) and FedScope (Federal Workforce Data). The sample period spans from 2020 to 2023. The FEVS, conducted by the Office of Personnel Management (OPM), targets permanent full- and part-time federal employees. The survey is distributed to all employees, with response rates ranging from 35% to 44% across years (OPM, 2023).

FedScope, managed and released by OPM, offers demographic and human resources data on federal agencies. The dataset provides de-identified individual-level data on age, education level, salary, years of service, supervisor status, employment type, and work unit information. Additionally, FedScope's online data cubes offer information on employee headcounts by race/ethnicity and gender at the agency level. This information is utilized to construct agency-level

variables on average characteristics and demographic composition by supervisor status. Data linkage between FEVS and FedScope is conducted at the agency level.

From 2020 to 2023, the FEVS collected responses from 2.1 million federal employees. Focusing on supervisory personnel—defined as first-line supervisors, managers, or senior leaders—455,218 respondents are retained. FEVS only includes agency identifiers when an agency has at least 300 respondents, and demographic information is masked when there are fewer than ten respondents in any agency-by-supervisor-status-by-gender-by-race cell. After excluding respondents without agency or race/ethnicity information, the final sample became 403,135 (12% drop). Table A1 lists the agencies included in the analysis.

Given that the key focus of this paper is the dynamics between one's own race and their agency's racial composition, it is necessary to exclude observations with missing agency or race information. However, one might be concerned that people of specific racial/ethnic groups are more likely to skip the race question than others, leading to an unrepresentative sample. To address this issue, this paper employs a post-stratification weighting technique that reweights the results to reflect the joint distribution of agency and race among federal supervisors. Appendix C presents details on reweighting techniques and examines the robustness of various weighting approaches.

3.2. Variables and Measurement

Dependent Variable. The dependent variable is supervisors' job satisfaction. Following previous studies (Choi, 2013, 2017; Fernandez & Moldogaziev, 2013; Wang & Brower, 2019), two measurement items are used to construct the variable on job satisfaction — "Considering everything, how satisfied are you with your job?" and "Considering everything, how satisfied are you with your organization?" Both items are measured with a 5-point Likert scale. The average of the two variables is used to construct the dependent variable.

Key Independent Variables. The primary independent variable is the proportion of subordinates of the same race/ethnicity in an agency. Respondents' race/ethnicity is measured with a 5-bin grouping—White, Black, Hispanic, Asian, and Others. For the first four groups, same-race subordinates are measured as nonsupervisory employees sharing the same race/ethnicity in the same agency in a given year. The "Others" group includes Native Hawaiian and Pacific Islander (NHPI), American Indian and Alaska Native (AIAN), and individuals of two or more races, with same-race subordinates defined as those belonging to these categories. Although this oversimplifies the heterogeneity of this racial group, they constitute less than 5% of the sample.

It is worth noting that this paper constructs racial congruence at the agency level rather than the dyadic level. As discussed in the hypothesis development section, the logic of relational demography theory and the burden of representation literature extends to the aggregate organizational level, and prior empirical work has applied similar measures at the agency level. Tokenism theory, moreover, inherently focuses on organizational-level numeric distribution rather than dyadic relationships. Nevertheless, readers should note that the findings reflect the consequences of agency-level compositional context and cannot be directly extrapolated to dyadic supervisor-subordinate interactions.

Moderation Variable. The moderation variable — the proportion of same-race colleagues — is measured in a similar way. For White, Black, Hispanic, and Asian, the variable is constructed as the proportion of same-race/ethnicity employees who also have supervisor status within the same agency in a given year. For the "Others" group, this variable is measured as the composition of colleagues who also belong to the group.

Mediation Variables. Two mediation variables are involved — role clarity and supervisor relationship. Role clarity refers to the extent to which individuals have a clear understanding of

their responsibilities and the expectations from their job positions (Jong, 2016; Katz & Kahn, 2015; Verlinden et al., 2023). Hence, supervisors' role clarity is measured with a single questionnaire item: "I know what is expected of me on the job," based on a 5-point Likert scale.

Previous studies that have utilized FEVS also adopt various items in measuring supervisor relationships (Harrington & Lee, 2015; Kim & Ko, 2014; Wang & Brower, 2019). The core argument of this paper concerns the notion that supervisors of color might struggle between different expectations from their subordinates and (higher hierarchy) supervisors. Therefore, this paper uses six questionnaire items that capture the concept of whether the supervisors feel supported, respected, and trusted by the higher-level leadership. All items are measured with a five-point Likert Scale, and the variable is constructed using the average of all variables. Details of the questionnaire items can be found in Table A2, Appendix A.

Control Variables. The model includes a varied set of control variables. Individual-level covariates consist of gender, race/ethnicity, age (under or above 40), years of experience (below ten years, 10–20 years, and above 20 years), and military service experience. Agency-level covariates encompass employee characteristics by supervisor status at the aggregate level, including age, years of education, years of service, log salary, log number of employees, and share of females. The model also controls for the overall diversity in the agency at both subordinate and supervisor levels, as previous studies suggest that workforce diversity affects employees' work-related attitudes (Pitts & Jarry, 2009; Choi, 2017; Hauret & Williams, 2020; Moon, 2018). Following previous studies (Pitts & Jarry, 2009; Hur & Strickland, 2015; Moon & Jung, 2018; Pitts, 2005), this paper uses the Blau index of heterogeneity to construct the diversity measurement.

A few observations have missing values in the dependent and mediation variables. Specifically, 1.2% of observations had missing values for job satisfaction, 1% for role clarity, and

3.1% for supervisor relationship. While the missingness is minimal, one might still be concerned about potential bias if the missing data are not random. This paper uses multiple imputation techniques to impute missing values (Rubin, 1987; White et al., 2011). To address missing values in the control variables, this paper includes a dummy variable indicating missingness without excluding such observations.

3.3. Summary Statistics

Table 1 presents summary statistics reweighted using post-stratification weights by supervisor, race, and agency composition. The sample comprises 68% White, 14% Black, 9% Hispanic, 4.3% Asian, and 3.8% Other. Appendix C compares the sample composition with the Federal supervisor population.

[Table 1 Inserted Here]

The top panel presents the summary statistics of the dependent and mediation variables. Black and Asian supervisors tend to report higher job satisfaction and role clarity. Conversely, Hispanics and supervisors of other races tend to report lower values for these variables. In terms of relationships with supervisors, Whites, Blacks, and Asians tend to report higher values compared to their counterparts of Hispanic and Other races. The middle panel of Table 1 reports the summary statistics for the key independent variables. White supervisors tend to have a higher proportion of same-race subordinates and colleagues, averaging 62% and 69%, respectively. In contrast, supervisors of color have 7% to 18% same-race subordinates and 5% to 16% same-race colleagues. The bottom panel explores the demographic characteristics across races/ethnicities. A higher proportion of supervisors of color are also female compared to White supervisors. Black supervisors tend to be older and have longer experience in the federal government than other races/ethnicities, and Asian supervisors tend to be less experienced.

3.4. Method

Identifying the effect of having a higher or lower proportion of same-race subordinates is challenging due to self-selection. For example, individuals may intentionally choose agencies with demographically homogeneous environments, while agencies with less diversity might purposefully promote managers of color who are adept at navigating such environments. To address these concerns, this paper utilizes a two-way fixed effects (TWFE) model, incorporating agency-by-race fixed effects and year fixed effects. The panel structure allows direct control for time-invariant agency characteristics such as organizational culture and working environment, while the year fixed effects adjust for temporal changes across the federal government such as shifts in diversity awareness. Previous literature indicates that the TWFE model can effectively account for both unobserved unit characteristics and temporal effects (Brüderl & Ludwig, 2015; Fernández-Val & Weidner, 2018; Lee & Yu, 2010; Sobel, 2012).

The TWFE model also addresses potential confounding from structural differences in same-race subordinate exposure across racial groups. Since White supervisors systematically work in agencies with much higher proportions of same-race subordinates, observed racially divergent patterns could otherwise reflect baseline exposure differences rather than genuine heterogeneity. The agency-by-race fixed effects ensure that estimates are based on within-race, within-agency variation across time, rather than direct comparisons between White and minority supervisors. This paper estimates the following equation:

$$Y_{ijkt} = \alpha_0 + \times \sum_{j=1}^{N} \beta_j \left(SameRaceSubordinates_{jkt} \times D_{ij} \right) + \gamma_1 SameRaceColleagues_{jkt} + \lambda X_{ijkt} + \kappa Z_{kt} + \theta_{jk} + \delta_t + \varepsilon_{ijkt} \quad (1)$$

where Y_{ijkt} is the outcomes of interest (job satisfaction) for individual i of race/ethnicity group j in agency k in year t. $SameRaceSubordinates_{jkt}$ is the proportion of subordinates of same

race/ethnicity for respondents of race/ethnicity group j in agency k in year t. $SameRaceColleagues_{jkt}$ is the proportion of colleagues (employees also holding supervisory status) of the same race/ethnicity for respondents of race/ethnicity group j in agency k in year t. Note that these two variables are measured at race-by-agency-by-year level. Individuals of the same race/ethnicity within the same agency of the given year will share the same values.

This paper includes the proportion of same-race colleagues in the analysis, as relational demography theory suggests that colleague demographics shape work-related attitudes, a pattern verified by prior empirical studies (Nielsen & Madsen, 2017; Stainback & Irvin, 2012; Wang & Brower, 2019). While the two compositional measures are highly correlated in the raw data ($r = 0.988$), this correlation reduces substantially to 0.477 after accounting for the fixed effects, with VIFs ranging from 1.14 to 3.91, well within accepted thresholds. Appendix B provides formal diagnostics and robustness checks that model each variable separately, confirming that the main conclusions remain consistent across alternative specifications.

In the equation, D_{ij} is a set of dummy variables indicating the race/ethnicity group of individual i. Note that the main term $SameRaceSubordinates_{jkt}$ is not included; therefore, no reference group is specified. Each β_j thus reflects the association between same-race subordinate composition and the outcome of interest for the supervisors of the race/ethnicity group j, which capturing the difference in outcomes between agencies where 100% versus 0% of subordinates share the same race/ethnicity.

This paper opts for this interaction-based specification rather than estimating separate models by race/ethnicity group to better disentangle the association between racial concordance and the outcomes of interest from the overall agency racial composition. Estimating separate

models would risk high collinearity between the independent variable and agency-level racial composition controls.

The model further controls for individual covariates (X_{ijkt}), as listed in Table 1, including gender, age, and past experience in the military and federal government. The specification also includes time-variant agency covariates (Z_{kt}) that denote the aggregate level of employee characteristics, including the average age, years of education, years of service, log salary, log number of employees, and the gender and racial composition of the agency in the given year. θ_{jk} is the agency-by-race fixed effect. δ_t is the year fixed effect. ε_{ijkt} is the error component.

The paper also examines the mediation roles of role clarity and supervisor relationships. Following Pituch and Stapleton (2008), this paper applies the product-of-coefficients method to calculate the mediation effect, where the mediation effect of variable Z is defined as the product of the coefficient of X's impact on Z and the coefficient of Z's impact on Y when controlling for X. The bootstrap approach is applied to estimate the standard error of that mediation effect. Previous literature suggests that the bootstrap method might produce a more reliable confidence interval of the mediation effect than other methods (MacKinnon et al., 2004).

Specifically, resampling 1,000 times within the agency-by-year-by-race cluster is performed. For each bootstrap sample, the two sets of models are estimated. The first examines the association between the independent variables and the mediation variables. The second estimates the effect of the mediation variables on the outcome variables. The standard error of the mediation effect is defined as the standard deviation of the 1,000 estimates from the bootstrap samples. To investigate the moderation effect of colleague composition, the following regression is estimated:

$$Y_{ijkt} = \alpha_0 + \beta_1 SameRaceSubordinates_{jkt} + \beta_2 SameRaceColleagues_{jkt} + \beta_3 SameRaceSubordinates_{jkt} \times SameRaceCoworkers_{jkt} + \lambda X_{ijkt} + \kappa Z_{kt} + \theta_{jk} + \delta_t + \varepsilon_{ijkt} \quad (3)$$

This equation introduces an interaction term between the proportions of same-race/ethnicity subordinates and colleagues. β_1 denotes the association between same-race/ethnicity subordinate composition and the outcome when there are 0% same-race/ethnicity co-workers at the supervisory level. β_2 denotes the association between same-race/ethnicity colleague composition and the outcome when there are 0% same-race/ethnicity subordinates. β_3 is the primary parameter of interest, which denotes the change in the association between same-race/ethnicity subordinates composition and the outcome when the proportion of same-race/ethnicity colleagues increases.

One issue regarding the empirical design is that the key independent variable is measured at the race-by-agency-by year level. Therefore, while the final sample has 403,135 observations, there are only $5(\text{races/ethnicities}) \times 32(agencies) \times 4(years) = 640$ data points for the independent variables. In addition, the respondents are nested within each agency, which might result in correlations in their error terms. To account for this, cluster standard errors at the agency-by-year level are used. Appendix B selected different cluster levels as well as utilized the multilevel modeling (MLM) and bootstrapping methods to compute standard errors. The results remain robust and consistent across these different approaches.

4. RESULTS

4.1. The Racially Divergent Patterns in Same-race Subordinate Composition and Supervisors' Job Satisfaction

Table 2 presents the impact of working in an agency with more subordinates of the same race/ethnicity on the job satisfaction of supervisors by racial group. The results suggest a positive association of same-race/ethnicity subordinates at the agency level with White supervisors' job satisfaction, but a negative association for supervisors of color (except Asian). Specifically, for White supervisors, having 10 percentage points (pps) more same-race/ethnicity subordinates in their agency is associated with an increase of 0.19 points in job satisfaction (Column 1; $p < 0.01$), corresponding to 0.18 standard deviations. Conversely, supervisors of color experience the opposite pattern. When their agency has 10 pps more same-race/ethnicity subordinates, Black supervisors show a 0.24-point drop in job satisfaction (0.24 SD; Column 2; $p < 0.05$). For Hispanic supervisors, the impact is a 0.5-point drop in job satisfaction (0.5 SD; Column 3; $p < 0.05$). Overall, the results support the racially divergent patterns proposed by H_1 and H_2 — White supervisors experience increased satisfaction when working in an agency with more same-race subordinates, while supervisors of color experience decreased satisfaction.

[Table 2 Inserted Here]

4.2. Mediation Analysis

This section explores the mechanism of the associations between subordinate composition and supervisors' work-related attitudes. Figure 2 presents the mediation analysis results in path diagrams, separated by White supervisors (Panel A) and non-White supervisors (Panel B).

[Figure 2 Inserted Here]

Figure 2(a) shows that for White supervisors, there is no significant relationship between same-race subordinates and supervisors' role clarity ($\beta = 0.145, p > 0.1$), but a significant positive coefficient on supervisors' relationships with their higher-level leadership ($\beta =$

$0.908, p < 0.05$). However, as the study does not propose a hypothesis regarding this relationship, the paper avoids overinterpreting the results.

In contrast, Figure 2(b) reveals that for non-White supervisors, having a higher proportion of same-race subordinates is associated with significantly lower role clarity ($\beta = -1.608, p < 0.05$) and poorer relationships with their higher-level leadership ($\beta = -1.866, p < 0.01$). Specifically, for supervisors of color, a 10 pps increase in same-race/ethnicity subordinates in the agency results in a 0.16 point decrease in role clarity and a 0.19 point decrease in supervisory relationships. The right side of both diagrams shows that role clarity and supervisor relationships are positively related to job satisfaction for both White and non-White supervisors, with similar magnitudes across groups. To calculate the mediation effect, this paper applies the product of coefficient method with the bootstrapping approach to compute standard errors. Both mediation pathways are statistically significant for non-White supervisors, supporting H_3 and H_4.

4.3. Moderation Analysis

This section examines the moderating effect of the demographic composition on supervisory status to test the hypotheses derived from the burden of representation and tokenism theory. Specifically, the estimation interacts the variable on subordinates' racial/ethnic composition with the colleagues' composition at the agency level. Table 3 reports the estimated coefficients from equation (2). Column (1) focuses on the impact on job satisfaction. The coefficient for same-race/ethnicity subordinates is negative ($\beta = -2.142, p < 0.01$), indicating that having more same-race/ethnicity subordinates tends to reduce job satisfaction when there are no same-race/ethnicity colleagues at the supervisory level.

[Table 3 Inserted Here]

The interaction term is positive ($\beta = 4.895, p < 0.01$), suggesting that as the proportion of same-race/ethnicity colleagues increases by 10 pps, the negative association between same-race/ethnicity subordinates and supervisors' job satisfaction diminishes by 0.5 points. When the same-race/ethnicity colleagues reach 44%, the initial negative association between same-race/ethnicity subordinates and supervisors' job satisfaction shifts to a positive association. Columns (2) and (3) examine the relationship between same-race/ethnicity subordinates composition and role clarity and supervisor relationship, revealing a similar pattern. Overall, the evidence supports H_5.

Figure 3 reports the marginal effects of having a higher proportion of same-race/ethnicity subordinates based on the share of same-race/ethnicity colleagues at the supervisory level. When the proportion of same-race/ethnicity colleagues is low (below 44%), a higher share of same-race/ethnicity subordinates is associated with a drop in job satisfaction. However, the relationship reverses when the proportion of same-race/ethnicity colleagues in the leadership reaches 44%. This pattern aligns with tokenism theory, suggesting that supervisors of color experience less burden from representing their racial/ethnic group as their representation in leadership increases. This finding highlights the importance of critical mass in mitigating the potential negative consequences of demographic representation.

[Figure 3 Inserted Here]

5. DISCUSSION AND CONCLUSION

This study utilizes multi-year data from the federal government to examine how the racial/ethnic composition of subordinates at the agency level affects supervisors' job satisfaction. The findings reveal a racially divergent pattern: White supervisors experience increased job

satisfaction when working in agencies with higher proportions of same-race subordinates, while supervisors of color experience decreased satisfaction.

For White supervisors, the positive association aligns with relational demography theory and echoes previous research showing that employees benefit from demographic similarity with supervisors (Grissom et al., 2012; Grissom & Keiser, 2011; Moon, 2018). For supervisors of color, working in agencies with higher proportions of same-race/ethnic subordinates is associated with lower job satisfaction. These findings are consistent with research on the burden of representation, which highlights that minority bureaucrats often encounter tensions between the expectations of the clients they represent and the organizational norms (Carroll et al., 2019; Habyarimana et al., 2007; Herbert, 1974). Although this paper cannot directly measure "burden," the evidence of a decrease in job satisfaction among supervisors of color provides evidence consistent with the burden of representation framework. The mediation analysis shows that reduced role clarity and deteriorated supervisor relationships are consistent pathways linking same-race subordinate composition to lower job satisfaction, further aligning with theoretical expectations that supervisors of color could face challenges stemming from discrepancies between their subordinates' expectations and those of their organizations.

This paper further explores the potential mechanisms underlying the association between subordinate composition and supervisors' welfare. The analysis suggests that the negative consequences of the drop in satisfaction are mediated by the level of role clarity and the quality of supervisor relationships. In particular, supervisor relationships emerge as the most potent mechanism, as a higher share of same-race/ethnic subordinates at the agency level is associated with worse relationships between the supervisors and their higher-level leaders, reflecting challenges in maintaining good relationships when facing role conflict. Role clarity also

significantly contributes, indicating that having more same-race/ethnic subordinates is associated with higher role ambiguity, resulting in conflicts arising from misaligned expectations.

Importantly, the findings discovered that the demographic makeup of colleagues at supervisory levels moderates these adverse associations. The negative association between same-race subordinates and supervisor satisfaction becomes most pronounced when supervisors have few racially similar peers in leadership positions. This negative association diminishes as representation increases, eventually turning positive when the supervisor's racial group constitutes at least 44% of leadership. These dynamics align with tokenism theory, which posits that minorities face heightened challenges in environments where they lack representation (Kanter, 1977a, 1977b; Moore, 2013; Stroshine & Brandl, 2011). Prior research on critical mass similarly indicates that negative tokenism effects decrease as minority representation increases (Jackson et al., 1995; Kalbfeld, 2019). This paper extends this literature by demonstrating how achieving sufficient representation among peer colleagues specifically mitigates the burden of representation experienced by minority supervisors.

The findings contribute to theory by empirically testing the potentially inconsistent theoretical expectations of two theories: relational demography theory (which predicts that racial congruence should improve well-being) and the burden of representation (which suggests that having more same-race subordinates might create additional tension that hurts well-being). The findings of this paper imply that the benefits of working with similar others may not be realized when this similarity also leads one group to have higher expectations of another, creating additional burdens and role conflict. This finding expands understanding of when relational demography theory fails to explain relationships between people.

This paper also contributes to the theory of burden of representation by connecting its consequences to job satisfaction and testing specific mediation variables. The findings demonstrate that conflict over various role expectations and tension arising from organizational leadership account for the negative experience of representing one's own community. The findings also demonstrate that whether one theory surpasses another depends on the organization's racial distribution. The paper connects critical mass theory to bridge the inconsistency between relational demography and burden of representation, illustrating when and how the burden of representation becomes a benefit.

These findings have important implications for diversity management in public organizations. While federal agencies have invested substantially in workforce diversification based on assumptions about employee satisfaction benefits, this research reveals that such assumptions may not universally apply. The results highlight the necessity of considering diversity's impact on supervisors themselves, not just subordinates. When representation simultaneously benefits both leaders and those they supervise, it presents an ideal organizational strategy. Conversely, if increased representation benefits subordinates while harming supervisors, it may compromise both supervisor well-being and organizational effectiveness. By illuminating the potential costs borne by leaders of color when representing those from similar backgrounds, the findings underscore the importance of developing strategies to alleviate these negative effects.

Building on these insights, the findings highlight that diversity needs to go beyond token representation to reduce negative impact on supervisors of color. This aligns with previous studies indicating that effective diversity management is reached when organizations invest in both recruiting and promoting underrepresented people and implementing diversity programs within the organization (Kulik & Roberson, 2008; Pitts et al., 2010; Soldan & Nankervis, 2014).

Specifically, the findings identify role clarity and supervisor relationships as key mediators of the association between racial composition and supervisory job satisfaction. Given these empirical insights, organizations could benefit from reducing role conflicts and improving relationships with underrepresented mid-level managers. Previous studies have identified several effective strategies, such as incorporating diversity goals into agencies' strategic plans, improving commitment to diversity at the leadership level, and establishing diversity training and mentoring opportunities (Kulik & Roberson, 2008; Wyatt-Nichol & Antwi-Boasiako, 2012; Hur & Strickland, 2015). These strategies might potentially address the role expectation conflicts and tensions between minority supervisors and organizational leadership.

While this paper finds that a critical mass threshold could turn the negative burden of representation into a benefit, the policy implications of this are normatively complex. This should not necessarily be interpreted as a numeric goal for the federal government to pursue, as quota systems face political backlash and concerns about discrimination. The findings also should not be interpreted as supporting a policy change to increase homogeneity among supervisors, as that has negative implications for employees and clients, and raises concerns about discrimination. Although the empirical results do not address the trade-offs between different policy responses, the findings suggest that the negative experience of supervisors of color stems from their isolation, driven by an imbalance in the numerical distribution. One implication is that human resource management policies could reduce these effects by empowering and supporting supervisors to alleviate the experience of tokenism.

Specifically, the results suggest some possible management policies. For example, human resource managers could clarify job expectations and mitigate role ambiguity by explicitly incorporating equity-related responsibilities into supervisory performance criteria. Establishing

vertical support structures—such as mentoring or sponsorship programs linking minority supervisors with senior leaders—can strengthen leadership relationships and reduce isolation (Bozeman & Feeney, 2009; Keller et al., 2020). Agencies may also foster peer-learning communities and cross-agency leadership cohorts to build a sense of belonging and psychological safety (Getha-Taylor, 2018). HR strategies could also focus on cultivating environments that allow diverse leaders to exercise discretion without disproportionate emotional or relational strain. Finally, integrating indicators of managerial well-being and inclusion climate into federal workforce assessments would enable agencies to evaluate whether diversity initiatives promote both equitable representation and sustainable supervisory satisfaction. The analysis does not address whether such efforts would be effective or desirable. However, such efforts are solutions that directly address the empirical findings about how the demographic congruence of employees affects supervisor attitudes.

When interpreting the results of this paper, several limitations regarding the level of analysis should be considered. First, due to data constraints, the current study could not measure subordinate composition at the immediate supervisory level. Instead, this paper measures subordinate composition at the agency level. Because subordinates and supervisors within the same agency do not necessarily interact frequently, this limitation restricts this paper's ability to generate direct insights into whether the observed patterns are driven by actual supervisor-subordinate interactions. Instead, the results could only be interpreted as reflecting how agency-level racial composition impacts the supervisors' welfare.

This measurement approach raises three related concerns. First, agency-level measurement provides only an imprecise proxy for the extent to which a supervisor is exposed to same-race subordinates in their actual work. Because the agency-level average aggregates across employees

who are more distant from the supervisor than their immediate reports, any association detected at this level would likely be attenuated compared to associations at the direct supervisory level. This introduces classical measurement error in the independent variable, producing downward bias (attenuation bias) in the estimates (Bound et al., 2001). The fact that this paper detects significant associations despite this imprecise measurement strengthens confidence that the true relationship—were the present study able to measure direct supervisory composition—would be even more substantial. Additionally, robustness checks using data with one level below the agency (see Appendix Table B4) also support the conclusions. Therefore, although the imprecision of the measurement may affect the estimates, this paper believes that the key conclusions would persist even with more precise measurements. Nevertheless, the lack of ideal measurement hinders this paper's ability to provide more direct evidence.

Second, using an organizational-level proxy rather than individual dyad measurement may obscure patterns that arise specifically from interpersonal interactions. Some theoretical frameworks this paper draws on, such as relational demography theory and the burden of representation, are grounded in the logic of interpersonal dynamics. Therefore, the findings can only offer evidence consistent with their predictions at the aggregate level, and may not fully capture the underlying mechanisms operating at the dyadic level. The results should be interpreted as aggregate-level patterns, which may reflect the accumulation of multiple interpersonal dynamics within the same agency, the broader organizational culture and climate, or a combination of both.

Finally, even at the aggregate level, the precise mechanisms linking same-race subordinate composition to supervisor job satisfaction remain partially obscured. The mediation variables (role clarity and supervisor relationships) capture broad attitudinal outcomes rather than the specific

processes theorized to drive them, such as advocacy pressure, role conflict from community expectations, or emotional labor. The agency-level subordinate composition may shape supervisors' experiences through direct supervisory interactions, organizational climate, diversity policies, or expectations for group representation. Hence, this study cannot fully disentangle these pathways. Future research with more granular data on supervisory interactions and direct measures of advocacy burden would help clarify the mechanisms underlying these aggregate associations.

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FIGURES

Figure 1: Theoretical Framework and Hypotheses

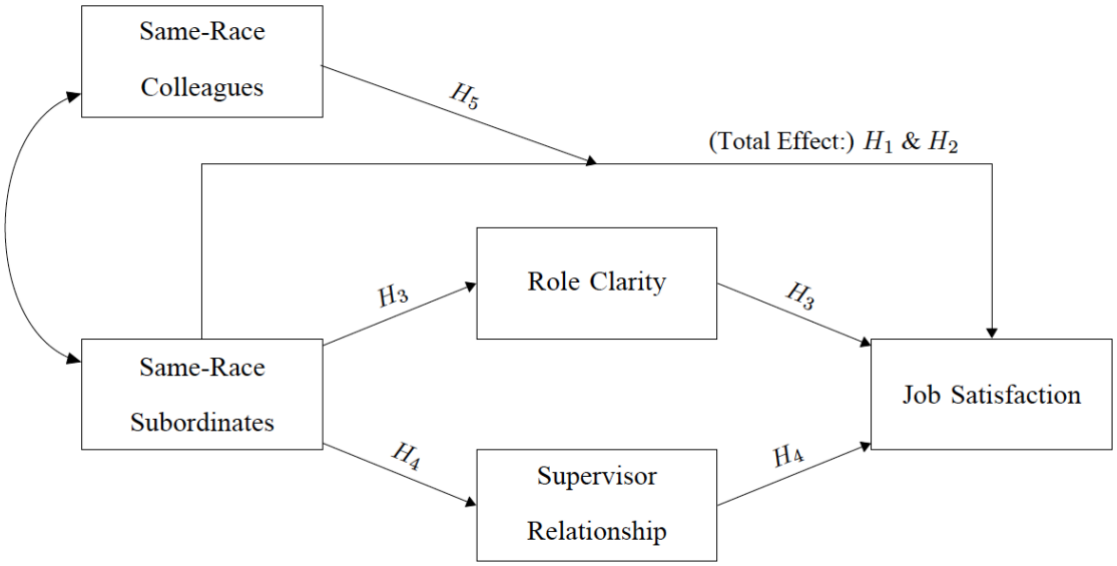
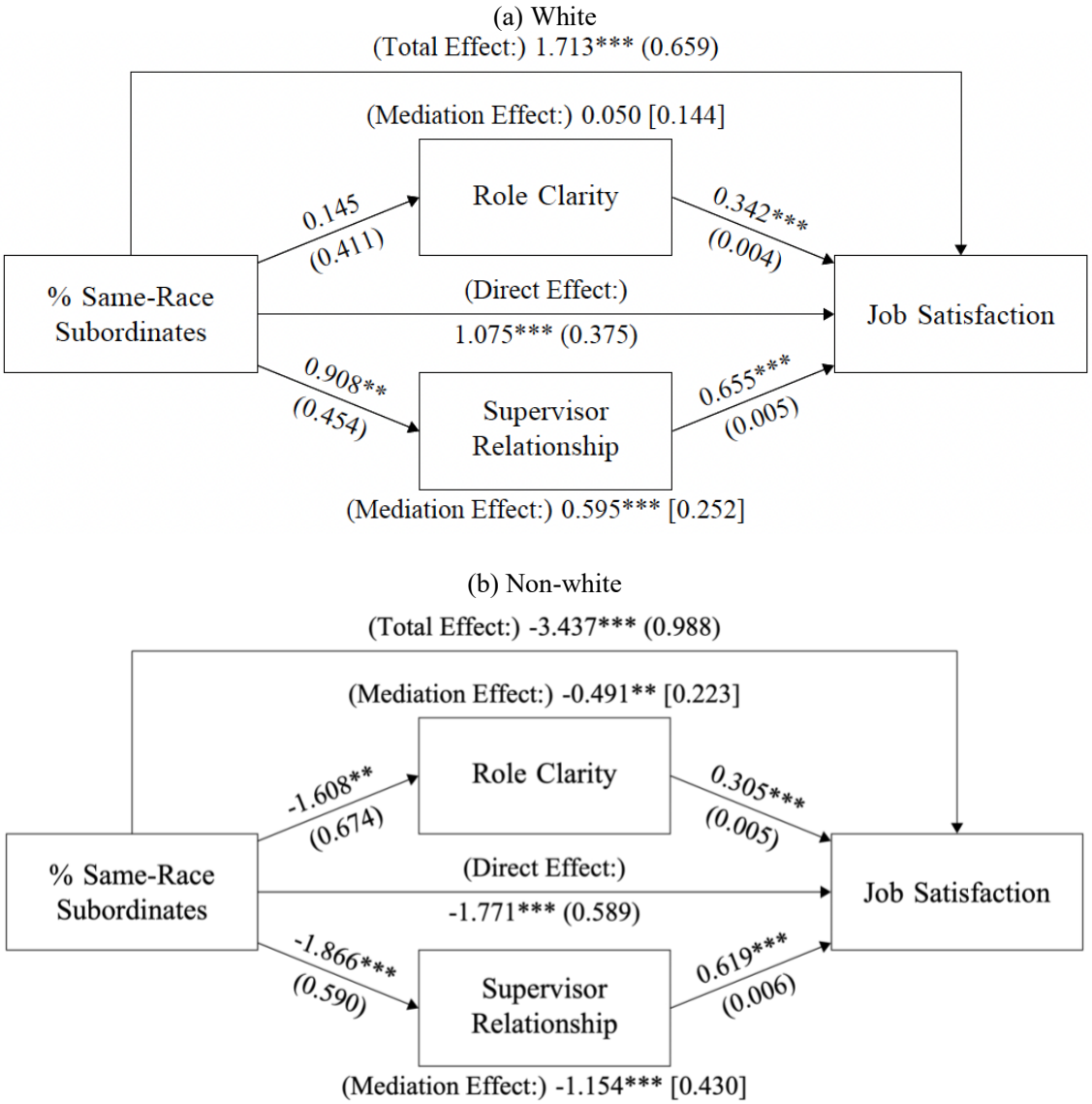
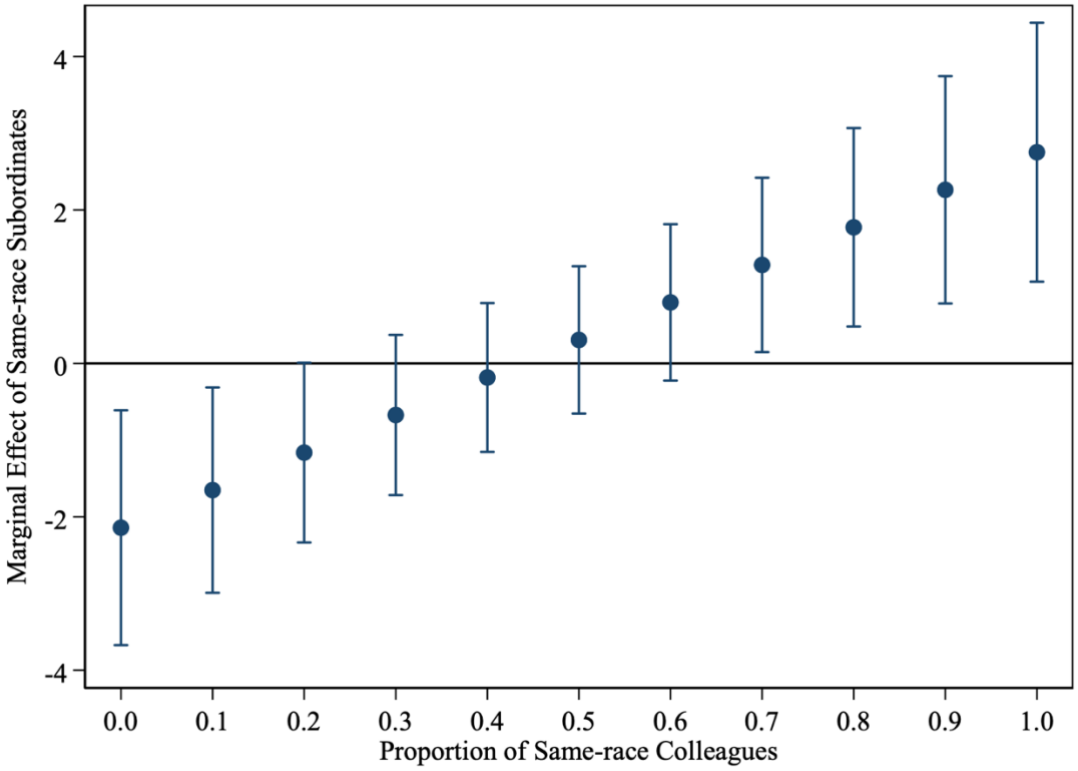


Figure 2: Mediation Analysis



Note: This figure summarizes the coefficients estimated from the mediation model. The mediation effect is estimated with the product-of-coefficients method. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard errors clustered at the agency by year level are reported in parentheses. Bootstrapping standard errors retrieved from 1,000 times resampling within the agency-by-year-by-race cluster in squared brackets. All models include individual and agency-level covariates, agency-by-race fixed effects, and year fixed effects. ***< 0.01, **< 0.05, *< 0.1

Figure 3: Moderate Effects of Proportion of Same-race Colleagues on Job Satisfaction



Note: This figure reports the marginal effects estimated from equation (2). All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. The horizontal axis denotes the proportion of same-race/ethnicity colleagues (i.e., who are also supervisors in the agency). The vertical axis denotes the marginal impact of same-race/ethnicity subordinates. The symbol denotes the point estimate, and the error bar shows the 95% CI, with standard error clusters at the agency by year level.

TABLES

Table 1: Summary Statistics

	All	White	Black	Hispanic	Asian	Others
Number of Observations	403,135	275,883	57,102	37,280	17,478	15,392
		(68.4%)	(14.2%)	(9.20%)	(4.30%)	(3.80%)
Dependent/Mediation Variables						
Job Satisfaction	3.86	3.85	3.95	3.83	3.94	3.70
	(1.03)	(1.03)	(1.04)	(1.05)	(0.91)	(0.95)
Role Clarity	4.18	4.17	4.25	4.17	4.26	4.07
	(0.92)	(0.91)	(0.97)	(0.95)	(0.83)	(0.87)
Supervisor Relationship	4.21	4.23	4.21	4.14	4.22	4.06
	(0.84)	(0.82)	(0.92)	(0.90)	(0.81)	(0.83)
Independent Variables						
% Same Race Subordinates	0.47	0.62	0.18	0.13	0.07	0.07
	(0.24)	(0.08)	(0.07)	(0.06)	(0.03)	(0.04)
% Same Race Colleagues	0.51	0.69	0.16	0.12	0.05	0.05
	(0.27)	(0.07)	(0.06)	(0.06)	(0.02)	(0.03)
Individual Demographics						
Female	0.37	0.33	0.53	0.37	0.40	0.39
	(0.48)	(0.47)	(0.54)	(0.48)	(0.48)	(0.44)
Age over 40	0.89	0.88	0.92	0.86	0.90	0.91
	(0.32)	(0.32)	(0.28)	(0.34)	(0.29)	(0.26)
Have Military Experience	0.36	0.37	0.38	0.34	0.15	0.37
	(0.48)	(0.48)	(0.52)	(0.47)	(0.35)	(0.43)
Years of Experience in Federal Government						
10 or fewer years	0.21	0.20	0.20	0.22	0.25	0.21
	(0.41)	(0.40)	(0.43)	(0.41)	(0.42)	(0.36)
10-20 years	0.43	0.43	0.40	0.44	0.44	0.43
	(0.50)	(0.49)	(0.53)	(0.49)	(0.49)	(0.44)
20 or more years	0.36	0.36	0.40	0.34	0.31	0.36
	(0.48)	(0.48)	(0.53)	(0.47)	(0.45)	(0.43)

Note: All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard deviation in parentheses. Job satisfaction, role clarity, and supervisor relationship are measured with a 5-point Likert scale and averaged across multiple items. The questionnaire items of these variables are available in Table A1.

Table 2: Associations between Agency-level Same-race/ethnicity Subordinate Composition and Supervisors' Job Satisfaction

	(1)	(2)	(3)	(4)	(5)
	White	Black	Hispanic	Asian	Others
% Same-Race Subordinates	1.922***	-2.362**	-4.966**	-2.604	-9.810**
$\times Race_j$	(0.676)	(1.062)	(1.982)	(2.660)	(3.980)
Observations			403,135		

Note: This table reports the coefficients estimated from equation (1). Each column reports the coefficient of β_j from the interaction term of $\%SameRaceSubordinates \times Race_j$. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard errors clustered at the agency by year level are reported in parentheses. The table only presents the coefficients of selected variables. All models include individual and agency-level covariates, agency-by-race fixed effects, and year fixed effects.
***< 0.01, **< 0.05, *< 0.1

Table 3: Moderation Analysis

	(1)	(2)	(3)
Dependent Variable:	Job Satisfaction	Role Clarity	Supervisor Relationship
% Same Race Subordinates	-2.142***	-1.155**	-1.181**
	(0.781)	(0.548)	(0.541)
% Same Race Colleagues	-1.568*	0.016	-0.422
	(0.896)	(0.616)	(0.615)
% Same Race Subordinates	4.895***	1.643*	2.661***
× % Same Race Colleagues	(1.319)	(0.903)	(0.841)
Turning point	0.438	0.703	0.444
Observations	403,135	403,135	403,135

Note: This table reports the coefficients estimated from equation (2). All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard errors clustered at the agency by year level are reported in parentheses. All models include individual and agency-level covariates, agency fixed effects, and year fixed effects.
***< 0.01, **< 0.05, *< 0.1

Appendix A: Data and Variables

Table A1: List of Agencies Included in the Sample

Code	Agency Name	2020	2021	2022	2023
Panel A: Cabinet Level Agencies					
AG	Department of Agriculture	✓	✓	✓	✓
CM	Department of Commerce	✓	✓	✓	✓
DD	Department of Defense	✓	✓	✓	✓
AF	Department of the Air Force	✓	✓	✓	✓
AR	Department of the Army	✓	✓	✓	✓
NV	Department of the Navy	✓	✓	✓	✓
ED	Department of Education	✓	✓	✓	✓
DN	Department of Energy	✓	✓	✓	✓
HE	Department of Health and Human Services	✓	✓	✓	✓
HS	Department of Homeland Security	✓	✓	✓	✓
HU	Department of Housing and Urban Development	✓	✓	✓	✓
DJ	Department of Justice	✓	✓	✓	✓
DL	Department of Labor	✓	✓	✓	✓
ST	Department of State	✓	✓	✓	✓
IN	Department of the Interior	✓	✓	✓	✓
TR	Department of the Treasury	✓	✓	✓	✓
TD	Department of Transportation	✓	✓	✓	✓
VA	Department of Veterans Affairs	N.P.	N.P.	N.P.	N.P.
Panel B: Large Independent Agencies (1000 or more employees)					
FQ	Court Services and Offender Supervision Agency for the District of Columbia	M	M	M	M
EP	Environmental Protection Agency	✓	✓	✓	✓
EE	Equal Employment Opportunity Commission	✓	✓	✓	✓
FC	Federal Communications Commission	M	M	M	M
FD	Federal Deposit Insurance Corporation	N.P.	N.P.	N.P.	N.P.
FR	Federal Reserve System	N.P.	N.P.	N.P.	N.P.
FT	Federal Trade Commission	M	M	✓	✓
GS	General Services Administration	✓	✓	✓	✓
LP	Government Printing Office	N.P.	N.P.	N.P.	N.P.
NN	National Aeronautics and Space Administration	✓	N.P.	N.P.	N.P.
NQ	National Archives and Records Administration	✓	✓	✓	✓
CU	National Credit Union Administration	✓	✓	✓	✓
NL	National Labor Relations Board	✓	M	M	M
NF	National Science Foundation	✓	✓	✓	✓
NU	Nuclear Regulatory Commission	✓	✓	✓	✓
OM	Office of Personnel Management	✓	✓	✓	✓
SE	Securities and Exchange Commission	N.P.	N.P.	N.P.	N.P.
SB	Small Business Administration	✓	✓	✓	✓
SM	Smithsonian Institution	N.P.	N.P.	N.P.	N.P.
SZ	Social Security Administration	✓	✓	✓	✓
IB	U.S. Agency for Global Media	M	✓	M	M
AM	U.S. International Development Cooperation Agency	✓	✓	✓	✓

Note: A "✓" indicates that the agency participated in the FEVS in the given year and is included in this study's analysis sample. "M" indicates that the agency participated in the FEVS, but the number of respondents was too small for OPM to release agency-specific identifiers, so the agency is excluded from the sample. "N.P." indicates that the agency did not participate in the FEVS in that year.

Table A2: Questionnaire Items of Variables

	Mean	SD	Cronbach's α
Job Satisfaction			0.899
Considering everything, how satisfied are you with your job?	3.924	1.055	
Considering everything, how satisfied are you with your organization?	3.791	1.088	
Role Clarity			
I know what is expected of me on the job.	4.182	0.910	
Supervisor Relationship			0.912
Supervisors in my work unit support employee development.	4.253	0.932	
My supervisor listens to what I have to say.	4.320	0.943	
My supervisor treats me with respect.	4.422	0.883	
I have trust and confidence in my supervisor.	4.216	1.063	
Overall, how good a job do you feel is being done by your immediate supervisor?	4.267	0.981	
I have a high level of respect for my organization's senior leaders.	3.779	1.170	

Note: All variables are measured with a 5-point Likert scale, with a value of 5 indicating "Very Satisfied" or "Strongly Agree" and a value of 1 indicating "Very Dissatisfied" or "Strongly Disagree." All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. This table was not estimated with multiple missing imputation.

Appendix B: Robustness Checks

The main analysis includes the racial group of "Others," consisting of AIAN, NHPI, and people of two or more races. However, this group is quite heterogeneous, raising the theoretical concern about whether AIAN supervisors could represent NHPI or any other similar combinations. Table B1 conducts a robustness check that excludes this group. The estimated results remain consistent.

Table B1: Robustness Check: Exclude "Others" Racial Group

	(1)	(2)	(3)
Dependent Variable:	Job Satisfaction	Role Clarity	Supervisor Relationship
Panel A: By Race/Ethnicity (Binary)			
% Same-Race Subordinates × White	1.862***	0.174	1.035**
	(0.686)	(0.424)	(0.470)
% Same-Race Subordinates × Non-White	-3.230***	-1.436**	-1.713***
	(0.992)	(0.658)	(0.613)
Panel B: By Race/Ethnicity			
% Same-Race Subordinates × White	2.052***	0.266	1.008**
	(0.707)	(0.408)	(0.483)
% Same-Race Subordinates × Black	-2.364**	-1.030	-1.836**
	(1.089)	(0.638)	(0.734)
% Same-Race Subordinates × Hispanic	-5.218***	-2.198	-1.432
	(1.961)	(1.506)	(1.087)
% Same-Race Subordinates × Asian	-2.861	-2.350	-1.751
	(2.704)	(1.439)	(2.547)
Panel C: Moderation Effect			
% Same-Race Subordinates	-1.944**	-0.992*	-0.984*
	(0.792)	(0.520)	(0.554)
% Same-Race Colleagues	-1.529	0.124	-0.384
	(0.933)	(0.624)	(0.616)
% Same-Race Subordinates	4.817***	1.450	2.533***
× % Same-Race Colleagues	(1.401)	(0.905)	(0.865)
Observations	383,811	383,811	383,811

Note: Panel A reports coefficients estimated from equation (1) and utilizes a binary indicator of race/ethnicity. Panel B reports coefficients estimated from equation (1) and utilizes a 5-bin grouping of race/ethnicity but excludes the other groups. Panel C reports coefficients estimated from equation (2). This table excludes respondents who are NHPI, AIAN, and two or more races. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard errors clustered at the agency by year level are reported in parentheses. All models include individual and agency-level covariates, agency fixed effects, and year fixed effects. ***< 0.01, **< 0.05, *< 0.1

The primary estimation controls for the agency's overall diversity using the Blau Index. While the Blau Index is commonly used in the literature (Hur & Strickland, 2015; Moon & Jung, 2018; Pitts, 2005), some argue that the entropy index or the proportion of minorities might be more appropriate (Choi, 2009, 2013, 2017; Choi & Rainey, 2010). Table B2 employs alternative diversity measurements to test for robustness, considering the case when using the Blau index for the entire agency (instead of subordinates and supervisors separately), the entropy index, and the composition of each race. The results remain robust across these alternative controls.

One concern for the analysis is that while the sample size is large, the variation of the independent variable (subordinates' racial composition) is measured at the agency-by-year level. In addition, respondents are nested within the agency-by-year clusters, leading to potential serial correlation in the error terms. The main analysis addresses this concern by clustering the standard errors at the agency-by-year level. Table B3 further considers other standard error computation approaches, including various cluster levels, a multilevel modeling (MLM) estimation, and bootstrapping standard errors. The results remain consistent.

Table B2: Robustness Check: Alternative Organization Diversity Control

	(1)	(2)	(3)
Dependent Variable:	Job Satisfaction	Role Clarity	Supervisor Relationship
Panel A: Control Overall Blau Index			
% Same-Race Subordinates × White	1.883***	0.284	0.836*
	(0.701)	(0.400)	(0.463)
% Same-Race Subordinates × Black	-2.308**	-0.868	-1.924***
	(1.032)	(0.618)	(0.682)
% Same-Race Subordinates × Hispanic	-4.865**	-2.076	-1.331
	(1.960)	(1.490)	(1.071)
% Same-Race Subordinates × Asian	-2.526	-2.353*	-1.627
	(2.657)	(1.415)	(2.563)
% Same-Race Subordinates × Others	-9.735**	-6.009**	-5.800***
	(3.965)	(2.936)	(2.002)
Panel B: Control Entropy Index			
% Same-Race Subordinates × White	2.212***	0.283	1.098***
	(0.539)	(0.373)	(0.369)
% Same-Race Subordinates × Black	-3.044***	-1.266*	-2.402***
	(1.114)	(0.645)	(0.724)
% Same-Race Subordinates × Hispanic	-5.426***	-2.286	-1.699*
	(1.875)	(1.488)	(0.979)
% Same-Race Subordinates × Asian	-3.023	-2.553*	-1.964
	(2.686)	(1.397)	(2.558)
% Same-Race Subordinates × Others	-10.241**	-6.114**	-6.155***
	(3.987)	(2.924)	(2.005)
Panel C: Control Share of Each Race/Ethnicity			
% Same-Race Subordinates × White	1.638***	0.116	0.846**
	(0.544)	(0.377)	(0.366)
% Same-Race Subordinates × Black	-3.494***	-1.261**	-2.466***
	(1.007)	(0.573)	(0.628)
% Same-Race Subordinates × Hispanic	-4.407**	-1.830	-1.148
	(1.713)	(1.434)	(0.954)
% Same-Race Subordinates × Asian	-2.437	-2.246	-2.891
	(2.693)	(1.383)	(2.479)
% Same-Race Subordinates × Others	-9.589**	-5.733*	-4.323*
	(3.868)	(2.989)	(2.286)
Observations	403,135	403,135	403,135

Note: This table reports the coefficients estimated from equation (1) but substitutes the covariates of the agency-level diversity index. Panel A utilizes the overall Blau index of the agency instead of separate indices for subordinates and supervisors. The formula of the Blau index is: $BI = 1 - \sum(p_i^2)$, where p_i stands for the proportion of the race group i in the measured population. In the case when the composition is dominated by only one racial group, the Blau index would be 0. Panel B utilizes the entropy index as computed as: $E = \sum_{j=1}^{N}(P_j \log_N(1/P_j))$. See Choi (2009) and Choi & Rainey (2010) for details. Panel C controls the proportion of black, Hispanic, Asian, and other racial groups for subordinates and supervisors separately. The proportion of white variables is ignored as the reference group. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard errors clustered at the agency by year level are reported in parentheses. All models include individual and agency-level covariates, agency fixed effects, and year fixed effects. ***< 0.01, **< 0.05, *< 0.1

Table B3: Robustness Check: Alternative Standard Error Estimation Methods

	(1)	(2)	(3)	(4)	(5)
		OLS		MLM	Bootstrap
Panel A: By Race/Ethnicity (Binary)					
% Same-Race Subordinates × White	1.713**	1.713**	1.713**	1.699***	1.713***
	(0.822)	(0.752)	(0.718)	(0.483)	(0.465)
% Same-Race Subordinates × Non-White	-3.437***	-3.437***	-3.437***	-3.288***	-3.437***
	(1.168)	(1.114)	(1.057)	(0.808)	(0.772)
Panel B: By Race/Ethnicity					
% Same-Race Subordinates × White	1.922**	1.922**	1.922***	1.717***	1.922***
	(0.804)	(0.754)	(0.712)	(0.488)	(0.476)
% Same-Race Subordinates × Black	-2.362**	-2.362*	-2.362**	-3.010**	-2.362**
	(1.089)	(1.237)	(1.185)	(1.064)	(0.997)
% Same-Race Subordinates × Hispanic	-4.966**	-4.966**	-4.966**	-3.453**	-4.966***
	(2.368)	(2.481)	(1.951)	(1.229)	(1.220)
% Same-Race Subordinates × Asian	-2.604	-2.604	-2.604	-1.079	-2.604
	(3.838)	(3.700)	(2.613)	(2.615)	(2.746)
% Same-Race Subordinates × Others	-9.810***	-9.810***	-9.810**	-6.517*	-9.810***
	(2.598)	(2.306)	(3.846)	(2.937)	(3.740)
Panel C: Moderation Effect					
% Same-Race Subordinates	-2.142**	-2.142**	-2.142**	-1.715**	-2.142***
	(0.979)	(1.047)	(0.915)	(0.661)	(0.692)
% Same-Race Colleagues	-1.568	-1.568	-1.568	-1.136	-1.568**
	(1.205)	(1.228)	(0.980)	(0.645)	(0.642)
% Same-Race Subordinates	4.895***	4.895***	4.895***	3.973***	4.895***
× % Same-Race Colleagues	(1.425)	(1.730)	(1.575)	(1.078)	(1.133)
SE Cluster Level	agency	agency-by-race	agency-by-race-by-year		
Random Effect				agency-by-year	
Observations	403,135	403,135	403,135	403,135	403,135

Note: Panel A reports coefficients estimated from equation (1) and utilizes a binary indicator of race/ethnicity. Panel B reports coefficients estimated from equation (1) and utilizes a 5-bin grouping of race/ethnicity but excludes the other groups. Panel C reports coefficients estimated from equation (2). Columns (1) to (4) estimate the model with OLS and cluster standard error at various levels (as denoted in the bottom panel of the table). Column (1) clusters standard error at the agency level; Column (2) utilizes a two-way cluster approach to cluster standard error at both the agency and year level; Column (3) clusters standard error at the agency-by-race level; Column (4) clusters standard error at the agency-by-race-by-year level. Column (5) utilizes the multilevel modeling (MLM) approach and includes the agency-by-year random effect. Column (6) computes standard error with the bootstrap approach, where this paper resampled 1,000 times with replacement within the agency-by-race-by-year cluster. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. ***< 0.01, **< 0.05, *< 0.1

A key limitation of this paper is that it relies on agency-level subordinate composition. However, people working in the same agency do not necessarily interact frequently. Table B4 conducts a robustness check using 2015 to 2019 FEVS data, which reports the work unit of one level below agencies. This data allows us to measure subordinate composition more closely. The limitation of the 2015 to 2019 data is that they only provide a binary indicator of minority status. Hence, this paper measures the "same-race" subordinates in a binary manner. That is, for a Black supervisor, all the non-White or Hispanic subordinates within the work unit are considered "same-race."

Table B4: Robustness Check: Using 2015 to 2019 Data

	(1)	(2)	(3)
Dependent Variable:	Job Satisfaction	Role Clarity	Supervisor Relationship
Panel A: By Race/Ethnicity (Binary)			
% Same-Race Subordinates × White	0.255***	0.126*	0.394***
	(0.094)	(0.067)	(0.070)
% Same-Race Subordinates × Non-White	-0.453***	-0.243***	-0.448***
	(0.102)	(0.072)	(0.074)
Panel B: Moderation Effect			
% Same-Race Subordinates	-0.561	-0.166	-0.681
	(1.204)	(0.692)	(0.743)
% Same-Race Colleagues	-1.146	-0.454	-1.099
	(1.311)	(0.740)	(0.784)
% Same-Race Subordinates	1.527	0.455	1.747
× % Same-Race Colleagues	(2.469)	(1.388)	(1.482)
Observations	404,099	404,099	404,099

Note: This table reports the results based on 2015 to 2019 FEVS and FedScope data. The proportion of same-race subordinates and colleagues is measured at one level below agency and using a binary indicator of minority status (White non-Hispanic, or otherwise). Panel A reports coefficients estimated from equation (1) and utilizes a binary indicator of race/ethnicity. Panel B reports coefficients estimated from equation (2). Standard errors clustered at the sub-agency by year level are reported in parentheses. All models include individual and sub-agency-level covariates, sub-agency fixed effects, and year fixed effects. ***< 0.01, **< 0.05, *< 0.1

Overall, the findings remain robust. Minority supervisors still report lower job satisfaction when having a higher share of same-race subordinates. When considering the moderating effect of colleague (at supervisor status) composition, the results demonstrate that when supervisors work in an environment with more same-race colleagues, the negative burden from same-race subordinates is reduced and eventually turns positive when they obtain a significant proportion. The estimates in this setting have smaller magnitudes than the main results, potentially due to the imprecise measurement of race/ethnicity. Still, the key findings remain valid.

One concern regarding the model specification is that the proportions of same-race subordinates and colleagues are highly correlated, potentially posing a threat of multicollinearity. However, as both variables are theoretically related to job satisfaction, omitting either might lead to omitted-variable bias. Therefore, in the main specification, this paper retains both variables.

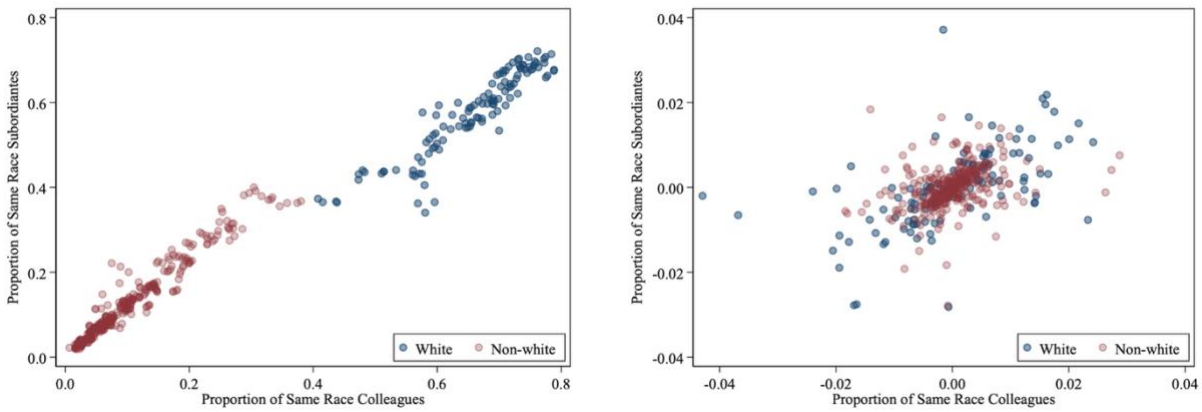
However, this paper relies on a TWFE model that controls for agency-by-race fixed effects and year fixed effects. The estimation only relies on the variation of the independent variable (proportion of same-race subordinates) within race-by-agency across years. The proportions of same-race subordinates and colleagues are highly correlated, particularly within the same agency; that is, agencies with more white supervisors tend to have more white subordinates. However, when the estimation relies only on the within-agency between-year variation, the two variables would not be as highly correlated as expected.

Figure B1a presents the scatter plot of the proportions of same-race subordinates and colleagues. As expected, the two are highly positively correlated ($r = 0.988$). However, Figure B1b shows a scatter plot that accounts for TWFE (i.e., demeaning each variable by the agency-by-race and year-level averages). The pattern no longer shows the strong correlation between the two variables (the partial correlation coefficient is 0.477). This paper also uses the demeaned version

variables (which is the same approach as the TWFE) to perform the regression and compute the variance inflation factors (VIF). For the main analysis (equation (1)), the VIFs for the proportion of same-race subordinates (and its interaction term with racial group) and for colleagues range from 1.14 to 3.91, which fall well within commonly accepted thresholds for multicollinearity.

Figure B1: Scatter Plot of Proportion of Same-race Subordinates and Same-race Colleagues

(a) Without TWFE (b) With TWFE



To further address this concern, Table B5 Columns (1) to (3) perform a robustness check that excludes the proportion of same-race colleagues from the model. The estimates from this alternative specification are largely aligned with the main findings.

In the main results, only the race variable is interacted with the composition of same-race subordinates when examining heterogeneous patterns. Table B5 Columns (4) to (6) further examine the results when also interacting the race variable with the composition of same-race colleagues. The overall pattern largely aligns with the main findings, despite the estimates becoming noisier, leading some to become insignificant. This could be due to multicollinearity inflating the standard errors. Nevertheless, the main conclusion remains consistent.

Table B5: Robustness Check: Alternative Specifications of % Same Race Colleagues

	(1)	(2)	(3)	(4)	(5)	(6)
Model Specification:	Exclude % Same Race Colleagues			Interact % Same Race Coll. w/ Race		
Dependent Variable:	Job Satisfaction	Role Clarity	Supervisor Relationship	Job Satisfaction	Role Clarity	Supervisor Relationship
Panel A: By Race/Ethnicity (Binary)						
% Same-Race Subordinates × White	2.209***	0.742***	1.521***	1.462*	-0.162	0.504
	(0.428)	(0.274)	(0.289)	(0.759)	(0.457)	(0.560)
% Same-Race Subordinates × Non-White	-3.091***	-1.191*	-1.438**	-3.072***	-1.162	-1.279**
	(1.024)	(0.672)	(0.591)	(1.057)	(0.742)	(0.639)
Panel B: By Race/Ethnicity						
% Same-Race Subordinates × White	2.413***	0.861***	1.508***	1.635**	0.083	0.243
	(0.456)	(0.263)	(0.321)	(0.768)	(0.499)	(0.530)
% Same-Race Subordinates × Black	-2.045*	-0.601	-1.457**	-1.842	-0.482	-0.980
	(1.121)	(0.645)	(0.725)	(1.168)	(0.777)	(0.853)
% Same-Race Subordinates × Hispanic	-4.569**	-1.702	-0.850	-5.221**	-2.968	-0.508
	(1.964)	(1.445)	(1.114)	(2.271)	(1.957)	(1.236)
% Same-Race Subordinates × Asian	-2.079	-1.782	-1.021	-3.554	-0.332	-5.895
	(2.601)	(1.496)	(2.566)	(4.741)	(2.457)	(4.547)
% Same-Race Subordinates × Others	-9.521**	-5.696*	-5.505***	-4.489	-4.246	0.940
	(3.973)	(2.950)	(2.017)	(6.854)	(4.179)	(3.462)
Observations	403,135	403,135	403,135	403,135	403,135	403,135

Note: Panel A reports coefficients estimated from equation (1) and utilizes a binary indicator of race/ethnicity. Panel B reports coefficients estimated from equation (1) and utilizes a 5-bin grouping of race/ethnicity but excludes the other groups. Columns (1) to (3) do not control for % same-race colleagues. Columns (4) to (6) further control for the interaction term between % same-race colleagues and racial groups. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard errors clustered at the agency by year level are reported in parentheses. All models include individual and agency-level covariates, agency fixed effects, and year fixed effects. ***< 0.01, **< 0.05, *< 0.1

In the main results, this paper utilizes multiple imputation to impute missing values in the dependent variables. Table B6 switches the approach to case-wise and list-wise deletion. The findings remain largely consistent.

Table B6: Robustness Check: Results Without Missing Imputations

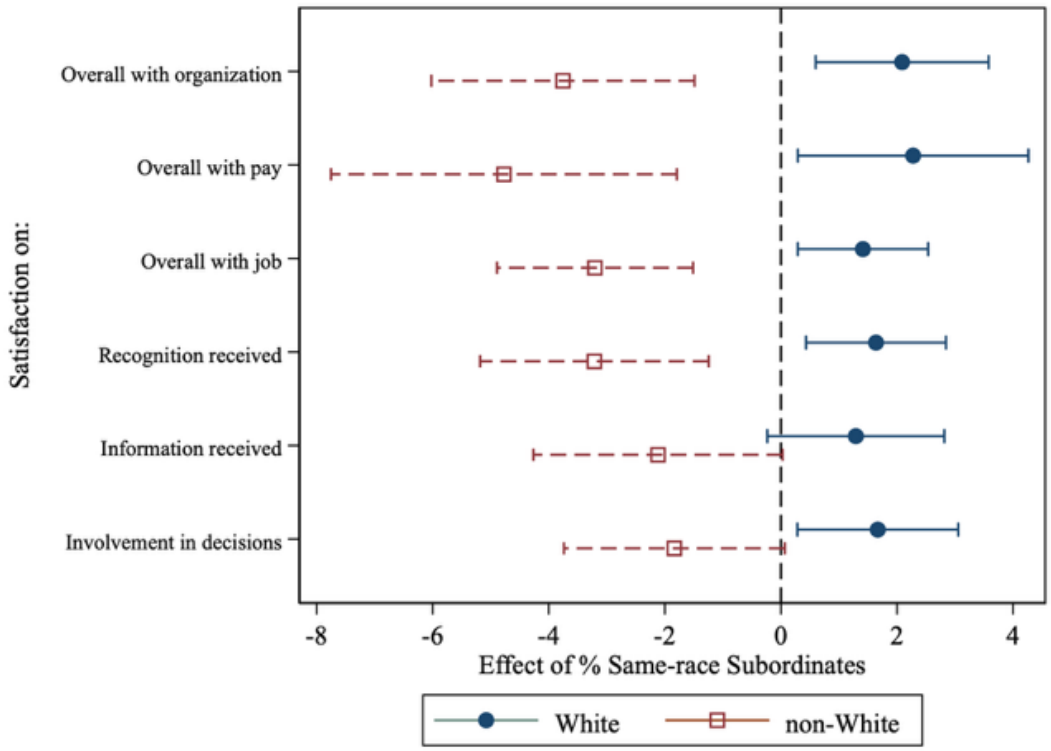
	(1)	(2)	(3)	(4)	(5)	(6)
Missing Values Handing:	Case-wise Deletetion			List-wise Deletetion		
Dependent Variable:	Job Satisfaction	Role Clarity	Supervisor Relationship	Job Satisfaction	Role Clarity	Supervisor Relationship
Panel A: By Race/Ethnicity (Binary)						
% Same-Race Subordinates × White	1.722**	0.115	0.903**	1.692**	0.060	0.837*
	(0.668)	(0.411)	(0.448)	(0.672)	(0.399)	(0.463)
% Same-Race Subordinates × Non-White	-3.490***	-1.698**	-1.819***	-3.518***	-1.735***	-1.858***
	(1.002)	(0.667)	(0.575)	(0.993)	(0.658)	(0.603)
Panel B: By Race/Ethnicity						
% Same-Race Subordinates × White	1.927***	0.235	0.895*	1.892***	0.135	0.818*
	(0.683)	(0.404)	(0.465)	(0.689)	(0.390)	(0.479)
% Same-Race Subordinates × Black	-2.428**	-1.098*	-1.863**	-2.491**	-1.368**	-1.960**
	(1.068)	(0.623)	(0.719)	(1.098)	(0.621)	(0.764)
% Same-Race Subordinates × Hispanic	-5.031**	-2.233	-1.123	-4.891**	-1.739	-1.162
	(2.003)	(1.494)	(1.076)	(2.032)	(1.467)	(1.071)
% Same-Race Subordinates × Asian	-2.621	-2.623*	-2.071	-3.393	-3.157**	-2.293
	(2.632)	(1.401)	(2.631)	(2.594)	(1.395)	(2.623)
% Same-Race Subordinates × Others	-9.582**	-5.869**	-5.406***	-9.287**	-5.465*	-4.452**
	(4.115)	(2.860)	(1.924)	(3.871)	(2.927)	(1.941)
Panel C: Moderation Effect						
% Same-Race Subordinates	-2.159***	-1.239**	-1.114**	-2.138***	-1.207**	-1.214**
	(0.794)	(0.547)	(0.520)	(0.778)	(0.535)	(0.545)
% Same-Race Colleagues	-1.551*	0.020	-0.340	-1.418	0.117	-0.305
	(0.893)	(0.610)	(0.613)	(0.899)	(0.584)	(0.595)
% Same-Race Subordinates	4.916***	1.718*	2.548***	4.822***	1.557*	2.633***
× % Same-Race Colleagues	(1.332)	(0.896)	(0.825)	(1.383)	(0.858)	(0.846)
Observations	398,353	399,128	390,476	382,478	382,478	382,478

Note: Panel A reports coefficients estimated from equation (1) and utilizes a binary indicator of race/ethnicity. Panel B reports coefficients estimated from equation (1) and utilizes a 5-bin grouping of race/ethnicity but excludes the other groups. Panel C reports coefficients estimated from equation (2). Columns (1) to (3) only include observations with non-missing values of the corresponding dependent variables. Columns (4) to (6) only include observations with non-missing values of all three dependent variables. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. Standard errors clustered at the agency by year level are reported in parentheses. All models include individual and agency-level covariates, agency fixed effects, and year fixed effects. ***< 0.01, **< 0.05, *< 0.1

In the main analysis, job satisfaction is constructed with the average of two questionnaire items. Figure B2 extends this analysis by showing the associations between same-race/ethnicity subordinates with different dimensions of satisfaction — job, organization, pay, information, recognition, and involvement. Overall, supervisors of color experience a negative association between same-race/ethnicity subordinates and satisfaction across all measurement items. As

demonstrated in Figure B2, the effect sizes of same-race/ethnicity subordinates remain consistent across various satisfaction measures. For all measurements, White supervisors experience a benefit from having more same-race subordinates, while supervisors of color undergo negative associations.

Figure B2: Associations between Same-race Subordinates and Job Satisfaction: By Measurement Items



Note: This figure reports the coefficients estimated from equation (1) and utilizes a binary indicator of race/ethnicity, and substitutes the dependent variable to alternative measurements of job satisfaction. All statistics are weighted by post-stratification weights that re-adjust the sample to represent the agency-by-race composition of supervisors in federal agencies. The symbol denotes the point estimate, and the error bar shows the 95% CI, with standard error clusters at the agency by year level.

Appendix C: Reweighting

While FEVS questionnaires were sent to all qualified federal employees, not all responded. Average response rates range from 35% to 44%, varying by year (OPM, 2023). Therefore, the sample might not be representative of federal employees. To address this concern, this paper adopted post-stratification weights based on a supervisory-status-by-race-by-agency distribution. In other words, the samples are reweighted to represent the joint distribution of supervisors by race/ethnicity (five-bin grouping) within each federal agency in the sample.

This paper utilizes the following formula:

$$W_{jkt} = \frac{N.Population_{jkt}}{N.Sample_{jkt}} \tag{C1}$$

Where the weights of respondents of a given race j in agency k in year t are constructed as the ratio of the population number of supervisors of the given race, agency, and year, divided by the sample number of supervisors of the given race, agency, and year. That is, when a given group is underrepresented in the sample, they receive higher weights.

Table C1 Column (1) reports the unweighted and weighted race distribution of supervisors in the sample. In the unweighted results, Black supervisors are slightly underrepresented, while "Other" groups are slightly overrepresented. However, this distribution is adjusted in the weighted statistics. These weight adjustments are based on the agency-by-race distribution (among supervisors). That is, the weights not only adjust for cases when a race is under- or over-represented, but also when an agency is under- or over-represented. For example, if one agency has a lower response rate, the respondents in that agency also receive higher weights. Thus, the reweighted results can be generalized to the federal supervisor population, considering the relative size of each agency. This paper uses this weighting design in the main analysis.

Table C1: Sample Race Composition by Different Weighting

	Weighted by				
	Race by Agency	Race within Agency	Race acr. Fed Gov	Orig. FEVS Weights	Unweighted
White	68.4%	67.3%	67.9%	65.0%	69.0%
Black	14.2%	14.5%	14.6%	13.2%	12.2%
Hispanic	9.2%	9.3%	9.1%	11.4%	9.4%
Asian	4.3%	4.6%	4.4%	4.9%	4.5%
Others	3.8%	4.3%	4.0%	5.5%	4.8%

Note: This table reports the race composition of the sample under various re-weighting approaches. Column (1) reports the composition based on post-stratification weighting by supervisor status, race, and agency. Column (2) reports the composition based on post-stratification weighting of supervisor status by race within each agency (but maintains the sample composition of each agency). Column (3) reports the race composition based on post-stratification weighting of supervisor status by race composition for the entire federal government (does not adjust the race composition within each agency). Column (4) reports the composition based on the original weights in the FEVS data as constructed by OPM (based on agency-by-gender-by-minority status of all employees and does not adjust for supervisor status). Column (5) reports the unweighted statistics.

This paper also considers an alternative weighting design. Table C1 Column (2) is based on post-stratification weights that only adjust for the distribution of race within each agency, but do not re-adjust the employee size of each agency.

In addition, the main weights consider the race-by-agency distribution among the agencies included in the analysis sample. However, some agencies were excluded from this study because they were too small, and their agency identifiers were masked in FEVS for confidentiality reasons. Column (3) of Table C1, therefore, considers an alternative weighting approach that is based on the race distribution (among supervisors) across the entire federal government. This takes into account the overall race composition of the federal government as a whole, including agencies not included in this study.

Finally, this paper also considers using the original weights from the FEVS study. OPM constructed FEVS sample weights using raking techniques based on population distributions by sex, agency, and minority status (OPM, 2023). The main limitation of their approach is that they

only consider minority status (White vs. non-White), instead of detailed race grouping. In addition, they only consider the race distribution of the agency, instead of the supervisory status-by-race distribution. However, there is a substantial difference in race distribution between supervisors and non-supervisors. For example, the proportion of White supervisors (among all supervisors) at federal agencies is around 68% in the 2020 to 2023 period. However, this proportion is only 60% among all employees. When using the original weights to re-weight the data, minorities might receive higher weights than they should (as the weights are based on overall race distribution, not distribution within supervisors).

Table C1 Column (4) demonstrates that results based on FEVS original weights lead to 65% White supervisors, which exaggerates the gap between the sample distribution and the population (68%). The FEVS original weights also did not successfully re-adjust the race distribution within minority respondents, as the weighting structure was only based on a simple binary indicator of minority status. However, to ensure robustness, this study still reports results based on the FEVS original weights.

Table C2 reports the robustness check of the estimated results across various weighting methods. Most estimates are very close to the main results. The estimates based on FEVS's original weights find a null effect on White supervisors but still a significant negative association on minorities.

Table C2: Reweighted Results

	(1)	(2)	(3)	(4)
	Weighted by			
	Race within Agency	Race acr. Fed Gov	Orig. FEVS Weights	Unweighted
Panel A: By Race/Ethnicity (Binary)				
% Same-Race Subordinates × White	1.830***	1.746***	1.073	1.769***
	(0.604)	(0.632)	(0.840)	(0.612)
% Same-Race Subordinates × Non-White	-3.321***	-2.869***	-2.812**	-3.170***
	(0.833)	(0.858)	(1.179)	(0.827)
Panel B: By Race/Ethnicity				
% Same-Race Subordinates × White	1.918***	1.844***	1.160	1.833***
	(0.603)	(0.637)	(0.832)	(0.609)
% Same-Race Subordinates × Black	-2.645**	-2.203**	-2.274	-2.634**
	(1.019)	(1.060)	(1.544)	(1.049)
% Same-Race Subordinates × Hispanic	-4.115**	-3.849**	-3.376*	-3.787**
	(1.728)	(1.696)	(2.003)	(1.696)
% Same-Race Subordinates × Asian	-1.000	-1.028	1.293	-0.486
	(2.359)	(2.271)	(2.523)	(2.230)
% Same-Race Subordinates × Others	-7.630**	-7.269**	-7.655**	-6.650**
	(3.194)	(3.317)	(3.668)	(3.142)
Panel C: Moderation Effect				
% Same-Race Subordinates	-1.698**	-1.203*	-1.791*	-1.210*
	(0.680)	(0.689)	(0.953)	(0.677)
% Same-Race Colleagues	-1.267	-0.919	-0.642	-0.976
	(0.843)	(0.868)	(1.091)	(0.842)
% Same-Race Subordinates	4.136***	3.390***	3.514**	3.331***
× % Same-Race Colleagues	(1.088)	(1.086)	(1.460)	(1.066)
Observations	403,135	403,135	403,135	403,135

Note: Panel A reports coefficients estimated from equation (1) and utilizes a binary indicator of race/ethnicity. Panel B reports coefficients estimated from equation (1) and utilizes a 5-bin grouping of race/ethnicity but excludes the other groups. Panel C reports coefficients estimated from equation (2). Column (1) reports the reweighted results based on post-stratification weighting of supervisor status by race within each agency (but maintains the sample composition of each agency). Column (2) reports the reweighted results based on post-stratification weighting of supervisor status by race composition for the entire federal government (does not adjust the race composition within each agency). Column (3) reports the reweighted results based on the original weights in the FEVS data as constructed by OPM (based on agency-by-gender-by-minority status of all employees and does not adjust for supervisor status). Column (4) reports the unweighted results. Standard errors clustered at the sub-agency by year level are reported in parentheses. All models include individual and sub-agency-level covariates, sub-agency fixed effects, and year fixed effects. ***< 0.01, **< 0.05, *< 0.1

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